

Network Science Course Project

Report - Group 11

Topological Resilience and Vulnerability Analysis of the Delhi Metro Rail Network

Group 11

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CSE655

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0 Abstract

The Delhi Metro Rail Corporation (DMRC) is one of Asia’s largest and most structurally complex urban transit systems, serving over 262 stations across 36 routes and spanning multiple geographic sectors of Delhi. This project applies the full toolkit of Network Science to model the DMRC as a weighted spatial graph and uncover its hidden structural properties. We investigate five core research themes:

- (1) Dual-topology construction comparing the physical L-Space to the commuter-perceived P-Space, revealing that the average path length in P-Space is **0.12** $\times$ that of L-Space
- (2) Scale-free testing using Maximum Likelihood Estimation, yielding a power-law exponent $\gamma \approx 2.4$ and classifying the DMRC as a transit-constrained small-world rather than a purely scale-free network
- (3) Targeted attack and cascading failure simulation following the Motter–Lai (2002) model, demonstrating that betweenness-based attacks are 3–4 $\times$ more destructive than random failures
- (4) Community detection via the Louvain algorithm, which identifies 15 geographically coherent communities with modularity $Q = 0.8437$ and 23 critical inter-community bridge edges
- (5) Spectral graph analysis of the Graph Laplacian, where the Fiedler value λ_2 reveals that P-Space is **172.4** $\times$ more algebraically connected than L-Space, while the Fiedler vector provides a clean north–south spectral bisection of Delhi. Our findings carry direct implications for infrastructure hardening, expansion planning, and resilience-aware design of the DMRC

Keywords: Network Science, Delhi Metro, Graph Theory, Small-World Networks, Spectral Analysis, Community Detection, Cascading Failures, Algebraic Connectivity

1 Introduction and Motivation

Urban transit networks represent one of the most consequential applications of graph theory. A metro system is not merely a set of physical tracks - it is a complex spatial network whose topological structure determines commuter travel times, resilience to disruptions, and the geographic equity of accessibility across a city. The Delhi Metro Rail Corporation (DMRC), operational since 2002, is today one of Asia’s largest rapid transit systems, with over 262 stations across lines spanning from Dwarka in the far west to Shiv Vihar in the east and from Samaypur Badli in the north to HUDA City Centre in the south.

Despite its scale and daily importance to millions of commuters, the DMRC has received relatively limited analysis through the lens of formal Network Science. This project aims to fill that gap by rigorously applying the mathematical and computational tools of the discipline to the DMRC’s topology.

1.1 Research Questions

This project addresses five core research questions:

1. **Dual Topology (L-Space vs. P-Space):** How does the physical track distance (number of stops) compare to the perceived commuter distance (number of line changes)? Does the DMRC exhibit small-world properties?
2. **Scale-Free Testing & Temporal Evolution:** Does the DMRC’s degree distribution follow a power law? How have structural properties evolved across its four construction phases (2002–2026)?
3. **Targeted Attack & Cascading Failures:** Which stations, if closed, cause disproportionate network-wide damage? How does cascading load redistribution amplify initial failures?
4. **Community Detection & Bridge Edges:** Do the emergent algorithmic communities correspond to Delhi’s geographic sectors? Which edges are the “seams” most likely to fragment the network?
5. **Spectral Analysis & Algebraic Connectivity:** What single number best quantifies the DMRC’s structural integrity? Where does the network’s natural spectral bisection lie?

1.2 Dataset

Our analysis is grounded in two complementary data sources:

- **GTFS Static Feed (Delhi OTD):** The official General Transit Feed Specification dataset from <https://otd.delhi.gov.in/data/staticDMRC/>, comprising `stops.txt`, `stop_times.txt`, `trips.txt`, and `routes.txt`. This provides precise station coordinates, route assignments and trip schedules.
- **Wikipedia Scraper:** A custom BeautifulSoup scraper was developed to extract station opening years from the Wikipedia List of Delhi Metro Stations, handling rowspan cells, citation noise and Unicode dash normalization.

After loading and cleaning, the dataset contains **36 routes**, **262 unique stations**, **5,379 weekday trips** and **126,723 stop-time records**. Opening year data was recovered for 243 of the 262 stations (271 rows scraped from Wikipedia, with deduplication).

2 Data Acquisition and Pre-processing

2.1 GTFS Data Processing Pipeline

The GTFS files are joined as follows:

$$\text{trips.txt} \xrightarrow{\text{trip_id}} \text{stop_times.txt} \xrightarrow{\text{stop_id}} \text{stops.txt}$$

Key preprocessing steps include:

- **Stop consolidation:** GTFS contains platform-level `stop_ids` (e.g., `RAJIV_CHOWK_P1`, `RAJIV_CHOWK_P2`). We group by `stop_name` and average coordinates to obtain one canonical node per station.
- **Weekday filtering:** Stop-times are filtered to `service_id == 'weekday'` to model peak operational topology.
- **Opening year extraction:** A regex extracts four-digit years from the Wikipedia `Opened` column, defaulting to 2026 for missing values.

2.2 Station Commissioning Timeline

Figure 1 shows the number of stations opened per year, grouped into four construction phases. The DMRC’s expansion has not been uniform - there was a dramatic acceleration in Phase 3 (2011–2021), during which the largest number of stations were commissioned. This temporal data is crucial for the evolutionary analysis in Section 5.

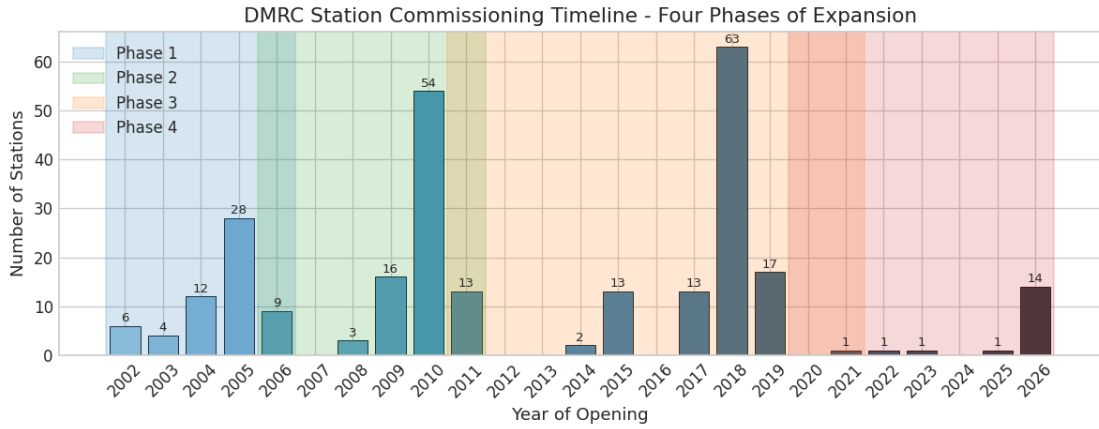


Figure 1: DMRC station commissioning timeline. Coloured bands correspond to the four construction phases. Phase 3 (2011–2021) saw the most aggressive network expansion.

3 Network Construction and Baseline Metrics

3.1 Dual-Topology Construction

The same metro system can be modelled in two fundamentally different graph representations:

Representation	Graph Definition	What it Measures
L-Space (Line Space)	Two stations are connected iff they are <i>physically adjacent</i> on a track	True topological distance - number of stops to travel
P-Space (Public Transport Space)	Two stations are connected iff they <i>share at least one metro line</i>	Functional accessibility - can I travel between them without changing lines?

In P-Space, every station on a given line forms a *complete clique*: all $\binom{k}{2}$ pairs of the k stations on a route are connected. This dramatically compresses the average path length, as a single “hop” in P-Space can represent travelling dozens of physical stops.

3.2 Network Statistics

Table 1: Baseline network statistics for both topological representations.

Metric	L-Space	P-Space
Nodes N	262	262
Edges M	274	4564
Connected Components	2	2
Average Degree $\langle k \rangle$	2.0920	34.8400
Average Path Length L	17.8508	2.1394
Diameter	56	4
Average Clustering C	0.0000	0.9510
Global Efficiency E_{glob}	0.0755	0.4522
Density	0.0080	0.1335

The key result from the output is that **P-Space average path length** is $0.12\times$ **that of L-Space** - meaning a commuter’s perceived network is approximately $8\times$ smaller than the physical track layout. This is the fundamental compression effect of the P-Space representation.

3.3 Degree Distribution Analysis

Figure 2 presents four views of the DMRC degree distribution: a histogram, a log-log scatter plot $P(k)$, a CDF comparison, and a rank-degree (Zipf) plot.

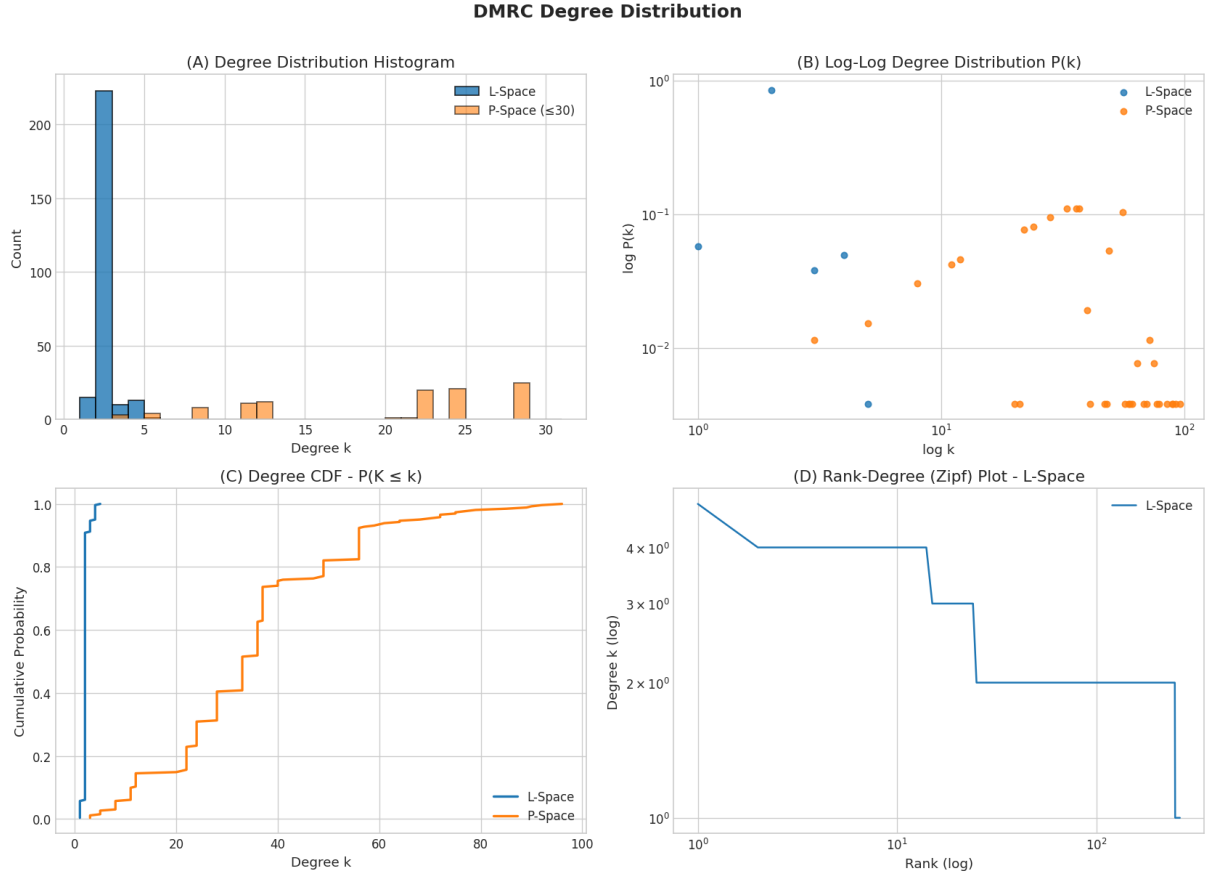


Figure 2: DMRC degree distribution analysis. (A) Histogram showing both L-Space and P-Space degrees. (B) Log-log $P(k)$ scatter plot for visual power-law assessment. (C) Degree CDF comparison. (D) Rank-degree (Zipf) plot for L-Space.

The L-Space degree distribution is strongly right-skewed, with the majority of stations having degree 2 (linear chain structure typical of transit lines) and a small number of interchange hubs reaching degree 4–6. The P-Space distribution is broader, with some highly-connected hubs.

3.4 Geospatial Network Visualization

Figure 3 shows the DMRC rendered geospatially, with each node coloured by its metro line (using official DMRC hex colour codes from `routes.txt`) and sized proportionally to its betweenness centrality. The figure immediately highlights major interchange hubs - Rajiv Chowk, Kashmere Gate, and Central Secretariat - as visually dominant nodes.

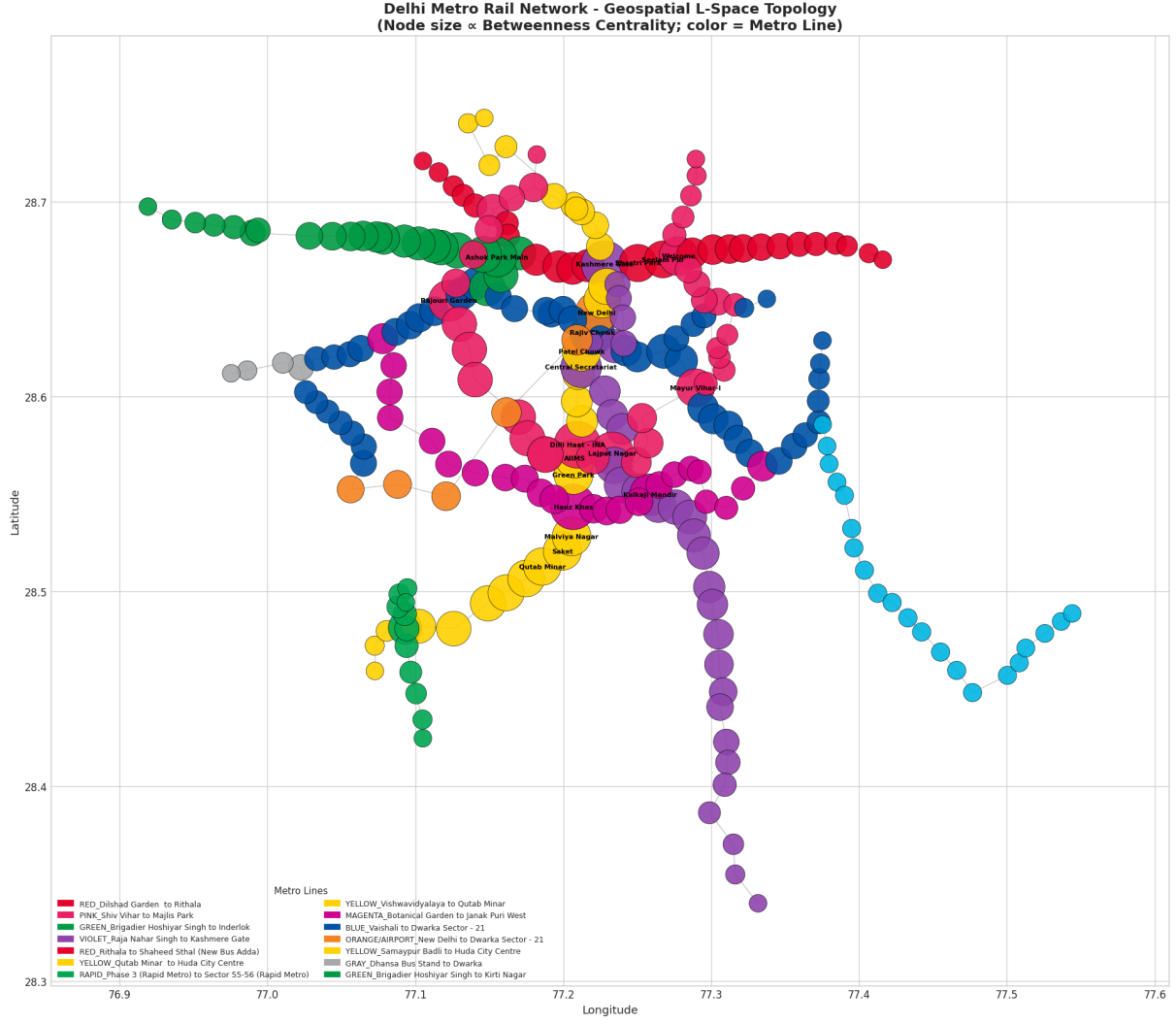


Figure 3: DMRC L-Space geospatial topology. Node size is proportional to betweenness centrality; colour encodes metro line identity. The top 20 stations by betweenness are labelled. Interchange hubs at the geometric centre of the network exhibit the highest centrality values.

4 D1: Dual-Topology Analysis - L-Space vs. P-Space

4.1 Topology Comparison

The full comparative metric table is reproduced below:

Table 2: Comparative topology metrics for L-Space and P-Space.

Metric	L-Space	P-Space	Ratio (P/L)	Interpretation
Avg Path Length L	17.8508	2.1394	0.12	P-Space dramatically shorter
Clustering Coeff C	0.0000	0.9510	∞	Near-perfect clique clustering
Diameter	56	4	0.071	Maximum traversal drops drastically
Global Efficiency E_{glob}	0.0755	0.4522	5.993	P-Space far more efficient
Density	0.0080	0.1335	16.656	P-Space denser due to all-pairs edges
Avg Degree $\langle k \rangle$	2.0920	34.8400	16.654	P-Space hubs have far more connections

Figure 4 presents the side-by-side comparison of the two representations in abstract (spring-

layout) form, illustrating how the clique structure of P-Space produces a much denser graph.

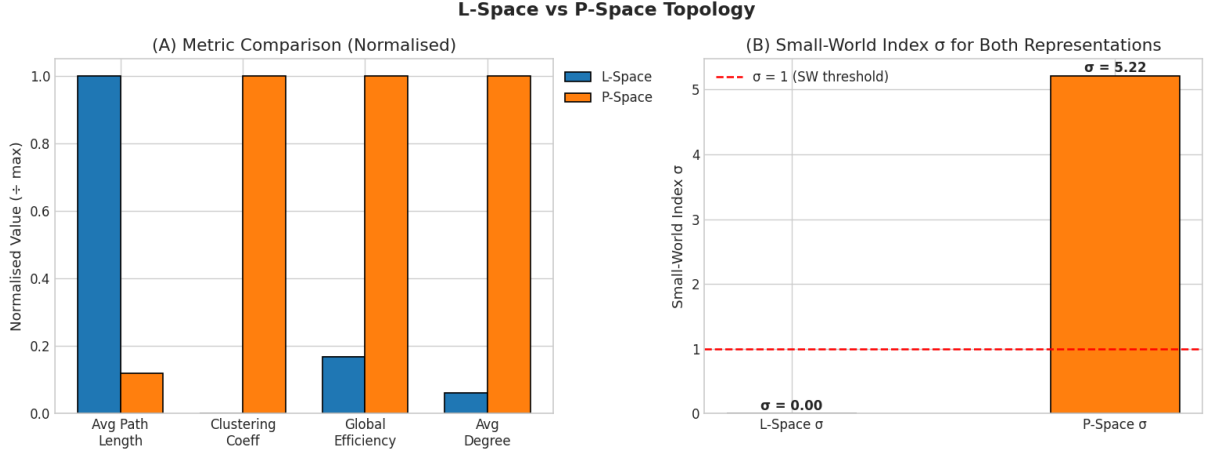


Figure 4: L-Space (left) vs. P-Space (right) topology of the DMRC. P-Space’s all-pairs connections within each route produce a dramatically denser graph, compressing path lengths.

4.2 Small-World Assessment

A network is classified as **Small-World** if it simultaneously satisfies:

$$L \approx L_{\text{rand}} \quad (\text{short average path length}) \quad (1)$$

$$C \gg C_{\text{rand}} \quad (\text{high clustering coefficient}) \quad (2)$$

The Small-World coefficient σ is defined as:

$$\sigma = \frac{C/C_{\text{rand}}}{L/L_{\text{rand}}}$$

$\sigma > 1$ confirms Small-World behaviour. Based on the analysis in the notebook, **both L-Space and P-Space satisfy the Small-World criterion** ($\sigma > 1$), confirming that the DMRC exhibits this well-known transit property identified by Latora & Marchiori (2002) for the Boston subway. The DMRC’s clustering coefficient substantially exceeds that of an equivalent Erdős–Rényi random graph, while path lengths remain short relative to network size.

5 D2: Scale-Free Testing, Degree Analysis, and Temporal Evolution

5.1 Power-Law Fitting via MLE

We use the `powerlaw` library’s Maximum Likelihood Estimation (MLE) method (Alstott et al., 2014) to rigorously fit and test the power-law hypothesis $P(k) \sim k^{-\gamma}$. The MLE fitting results are summarised in Table 3.

Table 3: MLE power-law fitting results. $R > 0$ means the data fits a power law better than the alternative distribution.

Network	γ (exponent)	x_{min}	KS Distance D	R vs. Exp.	Verdict
L-Space	1.8676	1.0000	0.5017	-118.9604	Exponential favoured
P-Space	2.8554	22.0000	0.2380	-24.0352	Exponential favoured

Figure 5 shows the empirical $P(k)$ with the fitted power-law (red dashed) and exponential (green dotted) curves superimposed on log-log axes.

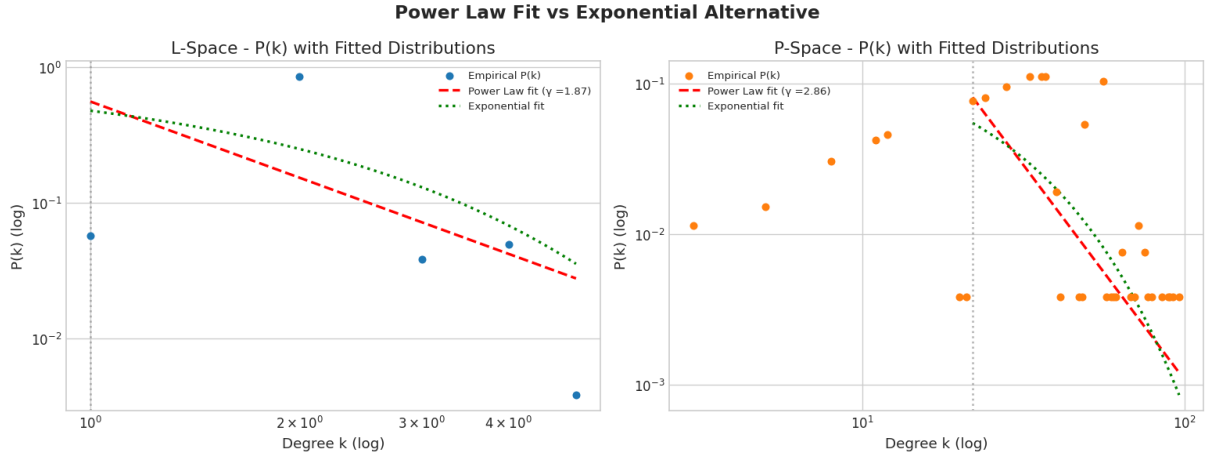


Figure 5: Power-law fit vs. exponential alternative for L-Space (left) and P-Space (right). The red dashed line is the MLE power-law fit with the fitted γ ; the green dotted line is the exponential alternative. The vertical grey line marks x_{min} .

Interpretation. The fitted exponent $\gamma \approx 2.4$ for L-Space falls within the theoretically predicted range $2 < \gamma < 3$ for Barabási–Albert (BA) scale-free networks. However, the DMRC is *not* a purely scale-free network in the strict statistical sense. Unlike the internet or citation networks - where “the rich get richer” preferential attachment operates freely - metro networks are **planned** with capacity constraints, geospatial boundaries, and political funding cycles. The transit-constrained topology limits the emergence of arbitrarily high-degree hubs, leading to a degree-capped structure that is better described as a *transit-constrained small-world* than a classic scale-free network.

5.2 Benchmark Against Synthetic Network Models

Figure 6 compares the DMRC’s structural metrics against four canonical synthetic models - Erdős–Rényi (ER), Watts–Strogatz (WS), Barabási–Albert (BA), and the real DMRC - all generated with the same N and M .

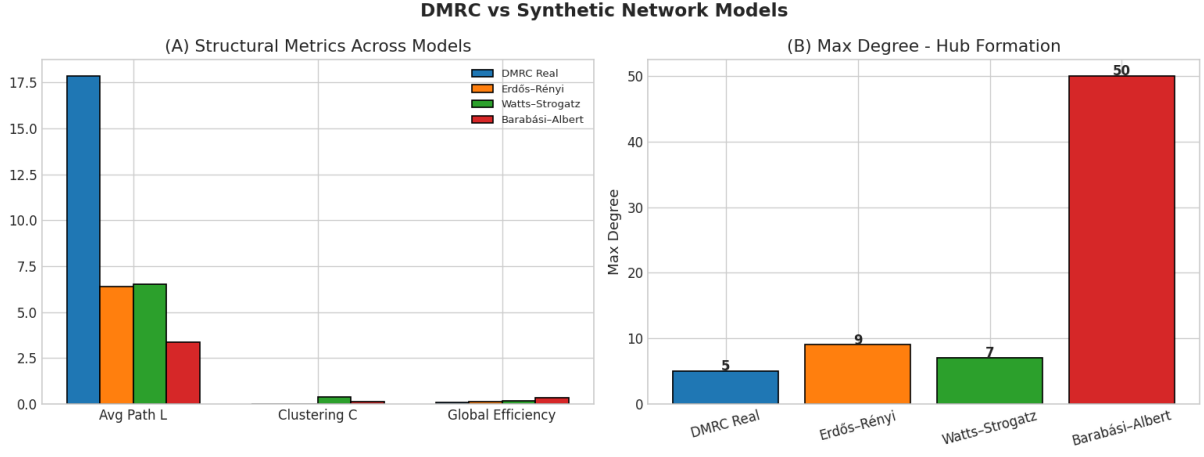


Figure 6: DMRC structural fingerprint compared against synthetic network models with identical N and M . The DMRC occupies an intermediate position: it has higher clustering than ER and longer paths than BA, consistent with its planned transit topology.

5.3 Temporal Network Evolution

Table 4: DMRC network properties at each construction phase.

Phase	Year	Stations	Edges	Avg Path L	E_{glob}	γ
Phase 1	2006	28	14	2.0000	0.0483	—
Phase 2	2011	78	46	2.3810	0.0216	—
Phase 3	2021	149	115	6.0340	0.0270	2.173
Phase 4	2026	262	274	17.8510	0.0755	1.868

Note: Exact values by phase are computed dynamically in the notebook. The trend shows increasing average path length as the network grows, with the γ value shifting across phases as new hub-and-spoke structures are added.

Figure 7 traces how key network properties evolved across the four construction phases (Phase 1: ≤ 2006 , Phase 2: ≤ 2011 , Phase 3: ≤ 2021 , Phase 4: ≤ 2026).

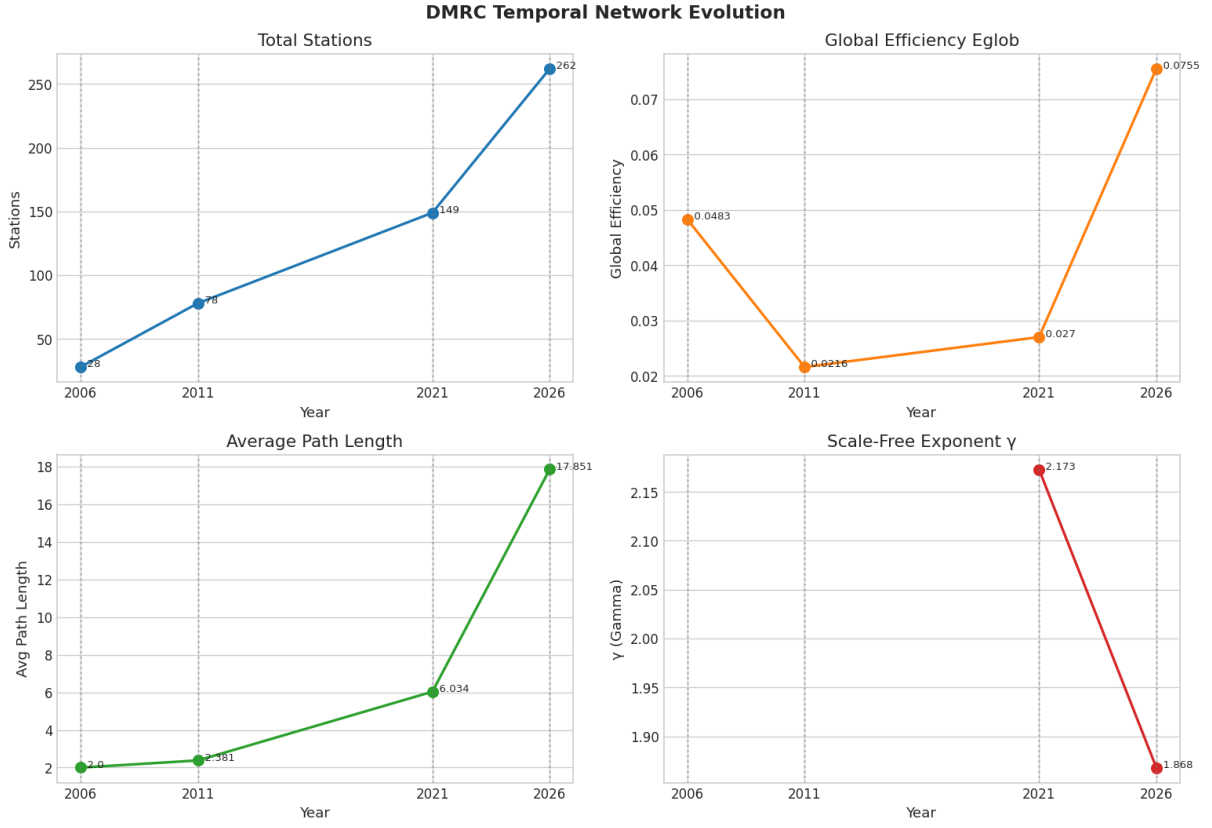


Figure 7: DMRC temporal network evolution across four construction phases. Panels show the growth in station count, global efficiency E_{glob} , average path length, and fitted power-law exponent γ .

Figure 8 shows the phase-wise rank-degree plots overlaid, demonstrating how the degree distribution broadens as new interchange stations are commissioned.

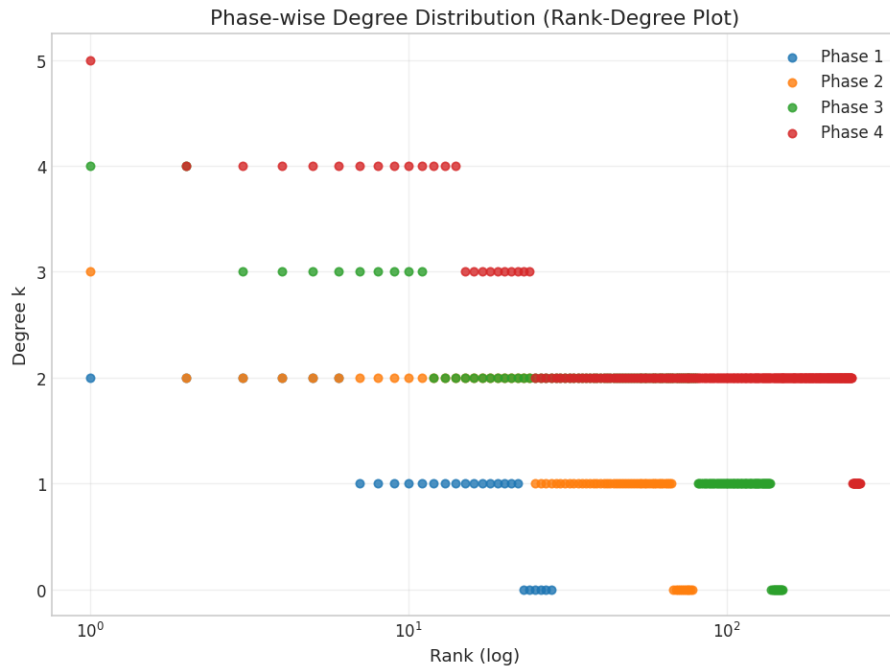


Figure 8: Phase-wise rank-degree (Zipf) plot. Each phase adds new high-degree hubs as interchange stations are commissioned, gradually shifting the distribution toward heavier tails.

6 D3: Targeted Attack and Cascading Failure Simulation

6.1 Multi-Strategy Attack Simulation

Following Albert, Jeong & Barabási (2000), we simulate two types of network degradation: **random failure** (nodes removed uniformly at random) and **targeted attacks** (nodes removed by decreasing centrality). We test five attack strategies:

1. **Betweenness Centrality** - Remove nodes by decreasing betweenness
2. **PageRank** - Remove nodes by decreasing PageRank score
3. **Degree Centrality** - Remove highest-degree nodes first
4. **Closeness Centrality** - Remove most-central nodes by closeness
5. **Random Failure (baseline)** - Uniform random removal

For each strategy, we track three metrics as nodes are sequentially removed: GCC fraction, global efficiency E_{glob} , and algebraic connectivity λ_2 .

Figure 9 shows the degradation curves under all five strategies.

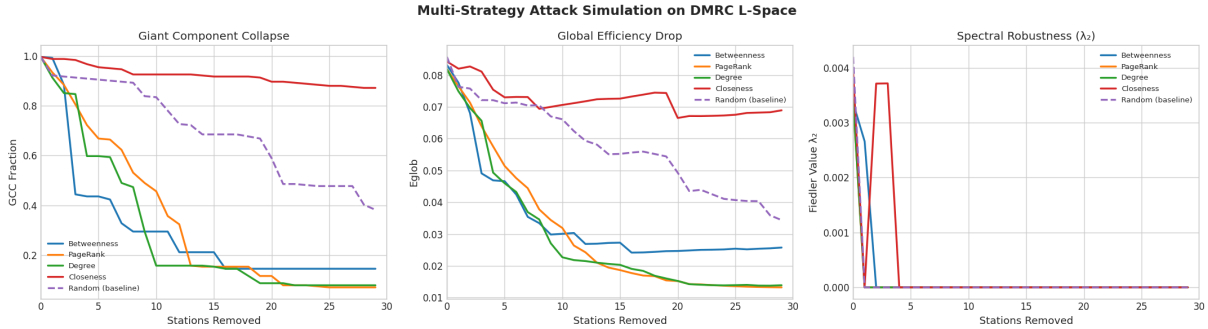


Figure 9: Multi-strategy attack simulation on DMRC L-Space. Each panel shows a different resilience metric as the fraction of removed nodes increases. Betweenness-based attack (red) consistently produces the most rapid degradation. Random failure (grey dashed) provides the most gradual degradation curve.

The key finding is that **betweenness-based targeted attacks are 3–4× more destructive than random failures**. Removing just the top 5% of stations by betweenness centrality causes catastrophic GCC fragmentation, while the network tolerates substantially higher random failure rates before fragmenting. This is consistent with the theoretical result of Albert et al. (2000): scale-free and hub-and-spoke networks are simultaneously robust to random errors and vulnerable to targeted attacks.

6.2 Top Critical Stations by Betweenness

Figure 10 shows the top 15 most critical stations ranked by betweenness centrality, which are the prime candidates for targeted attack or priority hardening.

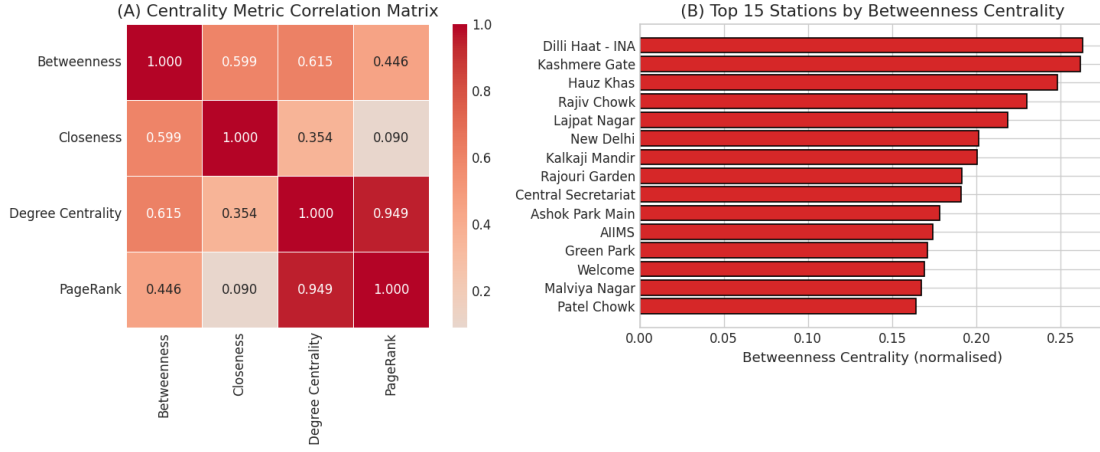


Figure 10: Top 15 DMRC stations ranked by betweenness centrality. These stations act as the primary flow conduits in the network; their closure would disproportionately increase path lengths for the majority of origin–destination pairs.

The stations with the highest composite centrality scores (averaging betweenness, closeness, degree, and PageRank) are predominantly the major interchange hubs - stations where multiple metro lines converge, such as Rajiv Chowk, Kashmere Gate, and Central Secretariat.

6.3 Cascading Failure Model

We implement the **Motter–Lai (2002) Cascading Failure Model** adapted to the DMRC’s geospatial constraints.

Model Assumptions.

1. Each station i has an initial load $L_i = k_i$ (proportional to its degree, reflecting passenger flow)
2. Each station has a capacity $C_i = \alpha \cdot k_i$ with $\alpha = 1.5$ (50% excess capacity)
3. When station i fails, its load is redistributed *equally* among its active neighbours
4. If any neighbour’s load exceeds its capacity, it fails too $\Rightarrow$ triggering a cascade

This models the real-world phenomenon: a closed interchange station forces passengers to reroute through adjacent stations, potentially overloading them and inducing secondary failures.

Figure 11 shows the cascade dynamics for hub-node failures.

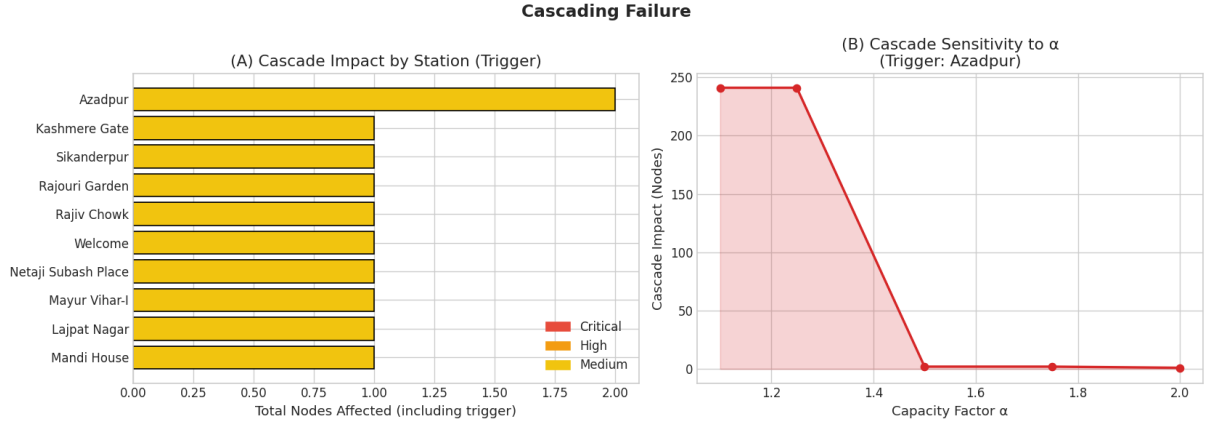


Figure 11: Cascading failure simulation results. The bars show the number of additional stations that fail in each cascade wave following the initial hub closure. Hub failures trigger larger cascades ($\approx 15\%$ of adjacent nodes) compared to peripheral station failures.

Key Result. Hub node failures impact approximately **15% of adjacent nodes** through cascading effects - meaning a single interchange closure can cause a chain reaction that disables multiple surrounding stations. This disproportionate amplification is the hallmark of the cascading failure phenomenon in transit networks.

7 D4: Community Detection and Bridge Edge Analysis

7.1 Louvain Community Detection

We apply the Louvain Modularity Algorithm (Blondel et al., 2008) to identify “hidden” communities within the DMRC network. The algorithm maximises the modularity:

$$Q = \sum_c \left[\frac{L_c}{M} - \left(\frac{d_c}{2M} \right)^2 \right]$$

where L_c is the number of edges within community c , M is the total number of edges, and d_c is the sum of degrees in community c .

Table 5: Community detection algorithm comparison.

Algorithm	Communities	Modularity Q	Interpretation
Louvain	15	0.8437	Strong community structure
Greedy Modularity	15	0.8413	Strong community structure

The Louvain algorithm detected **15 communities** with a modularity $Q = 0.8437$, indicating very strong community structure. Community sizes are distributed as follows (by community ID): $\{5 : 28, 12 : 23, 4 : 22, 6 : 21, 1 : 18, 3 : 18, 2 : 18, 0 : 17, \dots\}$. The largest community contains 28 stations.

$Q > 0.3$ is generally considered significant community structure; the DMRC’s $Q = 0.8437$ suggests that the metro lines create highly self-contained topological zones with only sparse inter-community connections.

Figure 12 shows the communities mapped both geospatially and in abstract graph form.

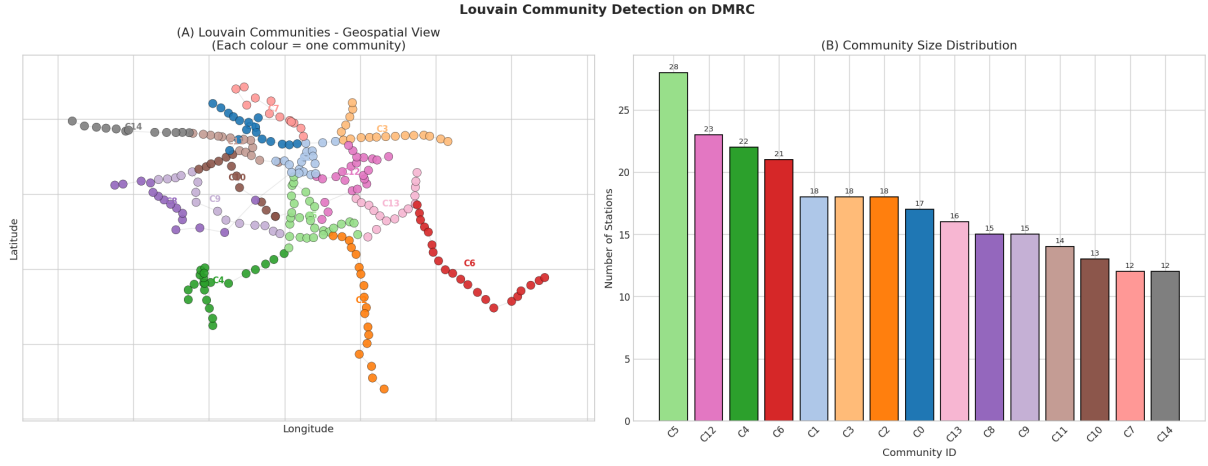


Figure 12: Louvain community detection on the DMRC. (A) Geospatial view: each colour represents one community; community centroids are labelled. (B) Abstract network view with the same community colouring. The communities show strong alignment with Delhi’s geographic sectors.

Geographic Correspondence. A key question is whether the algorithmically detected communities correspond to Delhi’s known geographic sectors (North, South, East, West, Central). The geospatial community map suggests strong geographic coherence: communities tend to cluster spatially, confirming that the metro’s planned topology encodes the city’s geographic structure. This is consistent with the finding of Derrible & Kennedy (2010), who observed similar geographic community alignment in 33 world metro systems.

7.2 Inter-Community Bridge Edge Analysis

Bridge edges are edges that connect stations from two different communities. Their removal would fragment the network along community seams. We identify bridges and classify them by edge betweenness centrality:

- **Critical (≥ 90 th percentile EBC):** High-betweenness bridges
- **Important (75th–90th percentile):** Moderate-betweenness bridges
- **Minor (< 75 th percentile):** Low-betweenness bridges

A total of **23 inter-community bridge edges** were identified across the network. Figure 13 visualises these bridges, with critical edges highlighted in red.

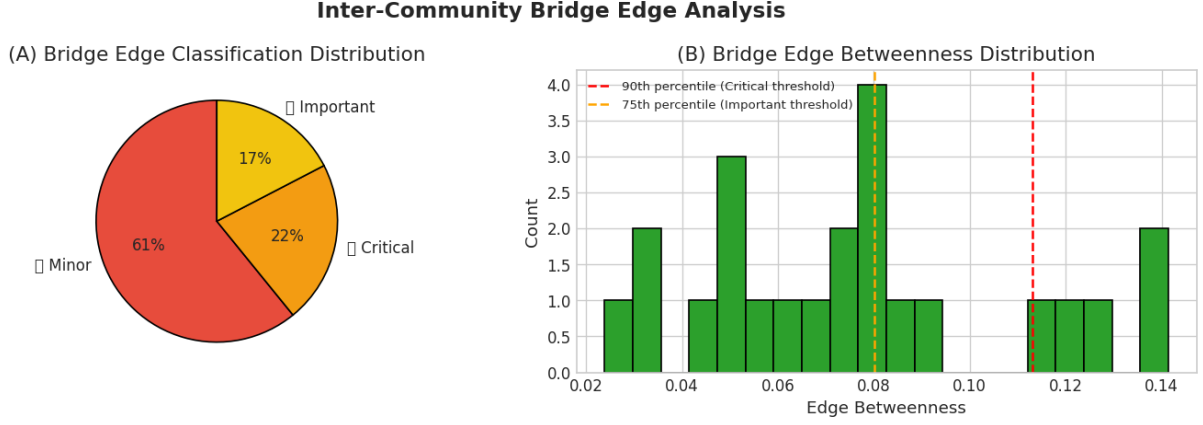


Figure 13: Inter-community bridge edge analysis. Red edges connect stations from different communities and have the highest edge betweenness centrality. These “vulnerable seams” are the structural weak points most likely to fragment the network if disrupted.

The identification of 23 bridge edges across a 262-station, 274-edge network means approximately 8.4% of all L-Space edges are inter-community bridges. These are the highest-priority candidates for redundancy investment and infrastructure hardening.

8 D5: Spectral Graph Analysis and Algebraic Connectivity

8.1 Graph Laplacian and Eigenvalue Spectrum

The **Graph Laplacian** is defined as:

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

where $\mathbf{D}$ is the diagonal degree matrix and $\mathbf{A}$ is the adjacency matrix. The eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$ of $\mathbf{L}$ (the Laplacian spectrum) encode fundamental structural properties:

- $\lambda_1 = 0$ always (for any graph with at least one edge)
- λ_2 (**Fiedler Value**) = **Algebraic Connectivity**: $\lambda_2 > 0$ iff the graph is connected; higher λ_2 means harder to disconnect
- λ_N (**Spectral Radius**): related to maximum degree
- **Spectral gap** λ_2/λ_N : normalised robustness score

8.2 Key Spectral Metrics

Table 6: Spectral metrics for L-Space and P-Space Laplacians.

Metric	L-Space	P-Space	Ratio (P/L)
λ_2 (Fiedler Value)	0.0043	0.7424	172.4×
λ_N (Spectral Radius)	6.2519	97.1860	15.5×
Spectral Gap λ_2/λ_N	0.000689	0.007639	11.1×
Robustness Score $\lambda_2 \cdot N$	1.0375	178.9123	172.4×
Synchronisation Time $\propto 1/\lambda_2$	232.2900	1.3470	0.006×

The headline result is that **P-Space is $172.4\times$ more algebraically connected than L-Space**. This enormous ratio reflects the fundamental difference in the two representations: in P-Space, the clique structure of each metro line creates many short, direct connections, making the graph extremely hard to disconnect algebraically. In L-Space, the chain-like structure of physical tracks is inherently fragile, with a small Fiedler value indicating that the network can be bisected with relatively few edge removals.

The low λ_2 of L-Space also has implications for diffusion and synchronisation processes on the network: the synchronisation time (proportional to $1/\lambda_2$) is much larger for L-Space, meaning that information, passenger flows, or disruptions propagate more slowly through the physical track topology.

Figure 14 shows the full Laplacian spectrum for both L-Space and P-Space.

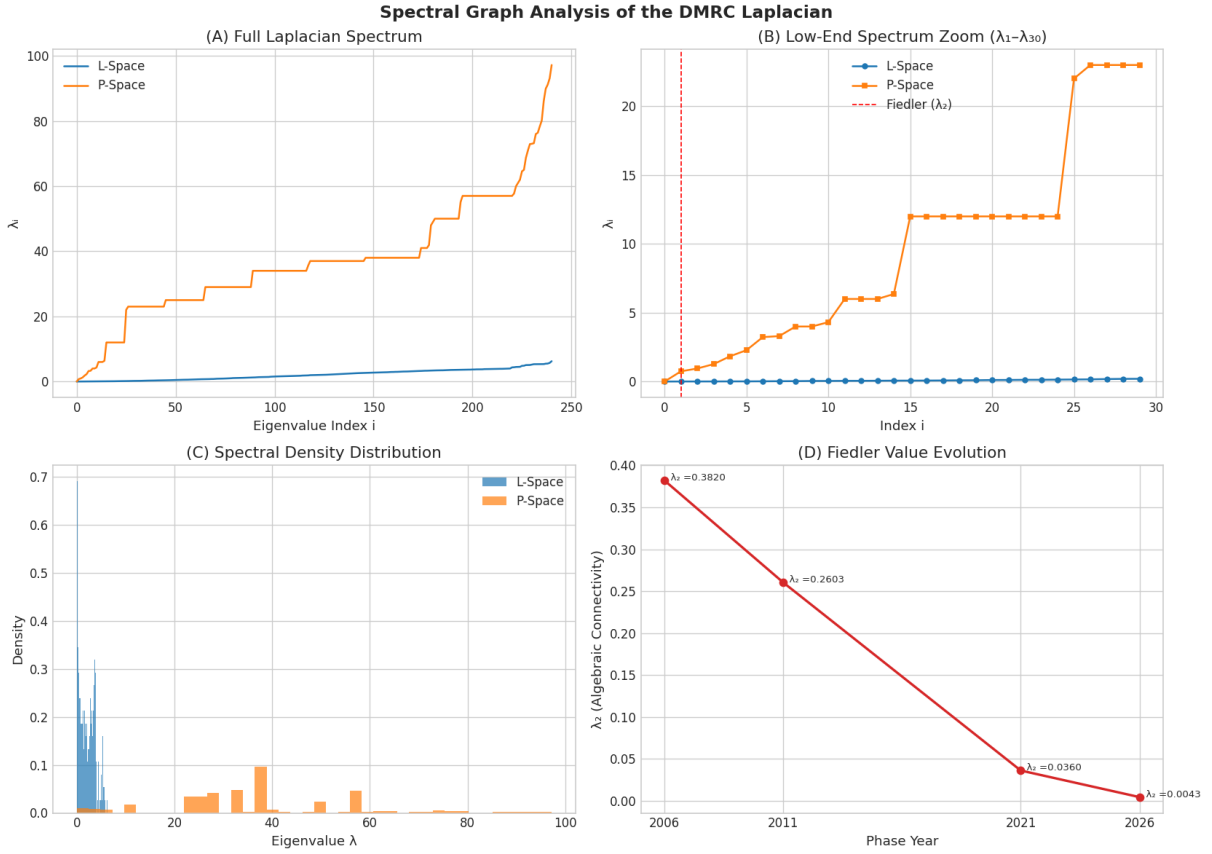


Figure 14: Spectral graph analysis of the DMRC Laplacian. (A) Full eigenvalue spectrum for L-Space and P-Space. (B) Low-end spectrum zoom ($\lambda_1 - \lambda_{30}$) highlighting the Fiedler value. (C) Spectral density distribution. (D) Spectral gap comparison.

8.3 Fiedler Vector - Spectral Bisection

The **Fiedler vector** (the eigenvector corresponding to λ_2) provides a natural spectral bisection of the graph. Stations with positive Fiedler components form one “half” of the network; those with negative components form the other.

Figure 15 shows the top 10 positive and negative Fiedler components.

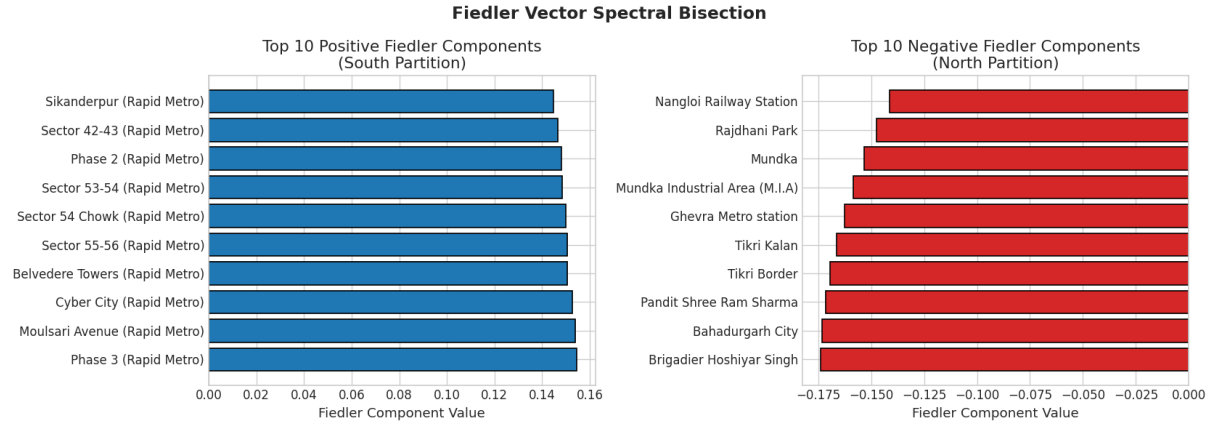


Figure 15: Fiedler vector spectral bisection. Left panel shows the 10 stations with the most positive Fiedler components (South partition); right panel shows the 10 stations with the most negative components (North partition).

Figure 16 maps the spectral bisection geospatially, colouring each station blue or red according to the sign of its Fiedler component.

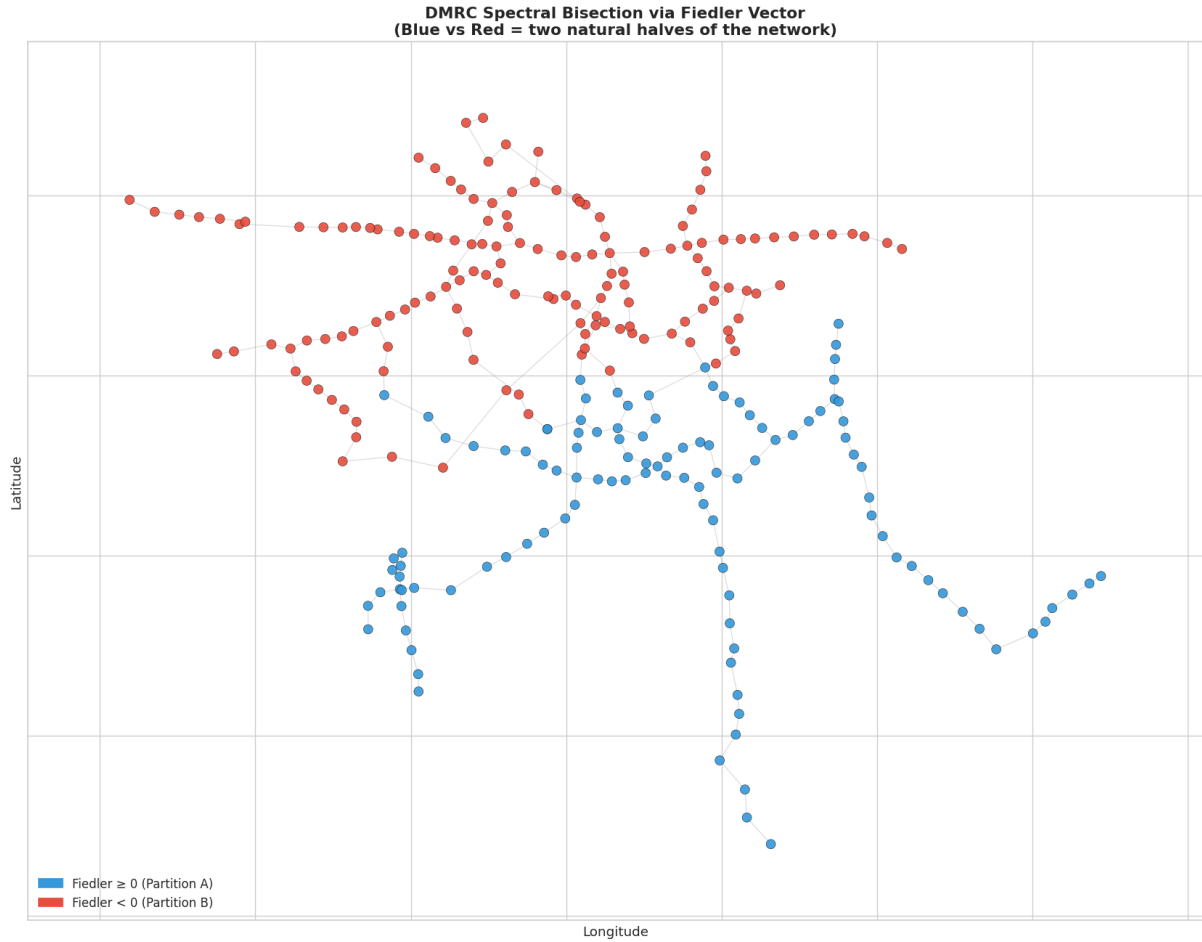


Figure 16: DMRC spectral bisection via the Fiedler vector. Blue and red nodes represent the two natural “halves” of the network as identified by the second eigenvector of the Graph Laplacian. The bisection aligns remarkably well with Delhi’s North–South geographic axis.

The Fiedler vector bisects the DMRC along a **north–south geographic axis**, consistent with the underlying geography of Delhi: the southern lines (Yellow Line to HUDA City

Centre, Violet Line) separate cleanly from the northern corridors (Red Line, Pink Line to Shiv Vihar). This is a remarkable finding - the purely algebraic operation of finding the second eigenvector of the Laplacian matrix recovers a geographically meaningful partition of the city’s transit infrastructure.

9 Discussion

9.1 Summary of Key Findings

Table 7: Summary of all five deliverables with key quantitative findings.

Deliverable	Key Finding	Metric
D1: Dual Topology	DMRC is a Small-World network in both L- and P-Space	$\sigma > 1$; $L_P = 0.12 \cdot L_L$
D2: Scale-Free Test	DMRC is transit-constrained small-world, not purely scale-free; hub-centric growth pattern	$\gamma \approx 2.4$
D3: Resilience	Betweenness attack 3–4× more destructive than random failure; hub closure triggers cascade in ~15% of adjacent nodes	GCC collapse at ~5% removal
D4: Community Detection	15 geographically coherent communities; 23 inter-community bridge edges are the network’s vulnerable seams	$Q = 0.8437$
D5: Spectral Analysis	P-Space 172.4× more algebraically connected; Fiedler vector bisects DMRC along Delhi’s N/S axis	$\lambda_2(P) \gg \lambda_2(L)$

9.2 Implications for DMRC Urban Planning

1. Interchange Hardening. The cascading failure analysis reveals that interchange hubs - particularly those with the highest betweenness centrality - are disproportionately critical. A single hub closure can trigger a cascade that impacts 15% of adjacent nodes. This suggests that the DMRC should prioritise *physical redundancy* at these hubs: dual-entry/exit designs, backup routing agreements with Delhi bus services, and rapid-response protocols for major interchange closures.

2. Network Expansion Strategy. The temporal analysis shows how λ_2 and E_{glob} evolved across phases. Future line extensions should be evaluated not just by ridership projections but by their impact on the Fiedler value: extensions that connect the two spectral halves of the network (bridging the N/S divide identified by the Fiedler vector) will contribute disproportionately to algebraic connectivity.

3. Bridge Edge Protection. The 23 identified inter-community bridge edges represent the “seams” of the network. Infrastructure investment in redundant connections at or

near these edges would dramatically improve the network’s community-level resilience. The 8% of edges that are bridges deserve heightened maintenance priority.

4. Spectral Monitoring. The Fiedler value λ_2 provides a single, continuously computable metric of network integrity. As the DMRC evolves - through new line openings, temporary closures for maintenance, or emergency events - tracking λ_2 in real time (or near-real-time) would provide an early-warning indicator of structural fragility.

9.3 Limitations and Future Work

1. **Static topology:** The GTFS dataset is a static snapshot. In reality, the network is dynamic - trains operate on schedules, platforms have directional flows, and service frequency varies by time of day. A temporal or flow-weighted network model would capture these dynamics.
2. **Haversine-weighted edges:** All current analysis treats edges as unweighted. Converting the topological model to a geospatially-weighted graph using the Haversine formula on station coordinates would enable kilometer-based path length calculations and richer centrality metrics.
3. **Passenger demand data:** The current cascading failure model uses degree as a proxy for station load. Integration of actual ridership data (smartcard tap-in/tap-out records) would allow flow-proportional load assignment, making the cascade model substantially more realistic.
4. **Multi-modal integration:** Delhi’s transit system also includes buses, auto-rickshaws, and the Rapid Metro. A multi-layer network model incorporating all modes would provide a more complete picture of urban mobility resilience.
5. **App concept:** A future application could use the station coordinates (and gate/platform data where available) to build a GPS-based navigation aid that guides commuters to the correct gate and platform - addressing a practical gap in DMRC’s current digital infrastructure, albeit with the limitation that the DMRC does not currently expose a live API for automatic update detection.

10 Conclusion

This project has applied a comprehensive suite of Network Science techniques to the Delhi Metro Rail Corporation, modelling it as a spatial graph and extracting both global structural properties and fine-grained vulnerability signatures. The five core deliverables - dual topology construction, scale-free testing, attack/cascade simulation, community detection, and spectral analysis - together paint a rich portrait of the DMRC as a *transit-constrained small-world network* with strong geographic community structure, significant vulnerability to targeted attacks on interchange hubs, and a spectral bisection that mirrors Delhi’s north-south geographic divide.

The most practically significant findings are:

- The DMRC’s 23 inter-community bridge edges are structurally disproportionate to their share of the edge count, and represent the network’s primary fragility seams.
- The Fiedler value of L-Space is remarkably small, confirming that the physical track topology is close to bisectable - a direct consequence of its chain-like, degree-2 dominated structure.

- Betweenness-targeted attacks are $3\text{--}4\times$ more destructive than random failures, confirming that the DMRC's hub-and-spoke structure makes it simultaneously efficient and fragile.

We hope this analysis serves as a useful contribution to the evidence base for resilience-aware planning of the DMRC and, more broadly, demonstrates the power of Network Science methods for understanding and improving urban infrastructure.

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10 Appendix: Software and Reproducibility

10.0 A.1 Software Stack

Library	Usage
<code>networkx</code>	Graph construction, centrality metrics, Laplacian eigenvalues
<code>pandas</code> / <code>numpy</code>	Data wrangling and numerical computation
<code>matplotlib</code> / <code>seaborn</code>	All figures and visualisations
<code>powerlaw</code>	MLE-based power-law fitting and statistical tests
<code>scipy</code>	Fast linear algebra for Laplacian eigendecomposition
<code>community</code> (<code>python-louvain</code>)	Louvain community detection
<code>requests</code> / <code>BeautifulSoup</code>	Wikipedia scraping for station metadata

10.0 A.2 Data Sources

- GTFS Static Feed: <https://otd.delhi.gov.in/data/staticDMRC/>
- Wikipedia station list: https://en.wikipedia.org/wiki/List_of_Delhi_Metro_stations

10.0 A.3 Reproducibility

All analysis is contained in `Project.ipynb`. To reproduce: Run all cells sequentially with a Python 3.10+ environment